



Fig. 2: SMD LED chips

LED AND OTHER LIGHTING SOURCES

Minimum light output (brightness)	Incandescent light bulbs	Common CFL bulbs	Common LED bulbs
Lumens	Energy used (watts)	Energy used (watts)	Energy used (watts)
450	40	9-11	5-8
800	60	13-20	10-13
1100	75	18-25	13-18
1600	100	23-30	20-26
2600	150	30-52	Currently not available

Courtesy: atcoenergysense.com

COB chips vary widely in their applications and thus different chips require different wattage, voltage and produce vastly different lumen counts. However, COB chips have good lumens per watt ratio, starting typically from 80 lumens per watt minimum and going up to 100 lumens per watt or more. This attribute of high efficiency makes these a clear favourite for all applications running on batteries, including mobile phones, cameras and more.

MCOB LEDs. A variant of COB is now gaining popularity—MCOB or multiple chip on board. MCOB devices (Fig. 4) are very similar to COB chips in their application and are essentially multiple COB chips together in series. MCOB devices produce a lot of light and are even better than COBs for low-wattage

applications. MCOB is a very new technology at this point that not many manufacturers are producing, but will likely catch on.



Fig. 3: COB LEDs

Why COB LEDs

From the above, it is clear that COBs are one of the best options available. These promise to lower the overall cost of LED luminaires by providing several benefits, such as greater design flexibility, better light distribution and simpler manufacturing process. Here is why:

1. COBs produce less heat as compared to other form of LEDs. This, in turn, results in higher efficiency. Better thermal dissipation is achieved because the LEDs are directly mounted on an aluminium sheet, which, in turn, offers better heat dissipation as compared to other materials.

2. COBs have a lower failure rate because spot soldering is no longer required as it is built directly on to the PCB.

3. Another major factor is that, typically COBs have a much wider beam angle and thereby a higher colour rendering index (CRI). This makes these a preferred choice for outdoor lighting luminaires such as streetlights.

4. It is also possible for COB LED package to be assembled in the heat-sink directly. In such a scenario, a separate mounting and PCB are not required.